

Early Outbreak: Estimating Disease Spread using Machine Learning

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1. Introduction

Early estimation of the basic reproduction number (R_0) is critical for guiding public health interventions during new outbreaks. R_0 is the number of secondary infections caused by a single individual in a fully susceptible population. A central question motivates this work: *How early in an outbreak can we accurately estimate R_0 from the observed epidemic trajectory using statistical and machine learning models?* Answering this question would allow for timely and effective responses to infectious disease threats.

To investigate, 1,500 deterministic SIR (Susceptible-Infected-Recovered) epidemic trajectories were simulated, and models were trained to predict R_0 from partial epidemic trajectories. We compare Linear Regression and Random Forest Regression across observation windows of 5, 10, 15, and 20 days, focusing on how predictive performance improves as more early outbreak information is available. The primary objective is to quantify the relationship between length of observation window and estimation accuracy.

This report presents initial findings from a project that will be extended into a 2-semester honors thesis, with future plans to incorporate stochastic simulations and integration of real-world outbreak data.

2. Methods

2.1 Susceptible-Infected-Recovered (SIR) Epidemic Model

The closed SIR model (no births or deaths) can be expressed by the following system of ordinary differential equations.

$$\frac{dS}{dt} = -\beta SI \quad (1)$$

$$\frac{dI}{dt} = \beta SI - \gamma I \quad (2)$$

$$\frac{dR}{dt} = \gamma I \quad (3)$$

where S is the proportion of susceptible individuals in a population, I is the proportion of infected individuals, R is the proportion of recovered individuals, β is the infection rate (product of contact rates and transmission probability), and γ is the recovery rate. $1/\gamma$ determines the average infectious period for an individual. These equations have initial conditions $S(0) > 0$, $I(0) > 0$, and $R(0) = 0$. Further, $S + I + R = 1$.

The following assumptions were made in the forming of these equations: First, every individual in a population has an equal probability of coming into contact with any other individual in the population. This is known as homogenous mixing. Further, all individuals have an equal probability of contracting the disease and passing on the disease. Demographics like age, sex and location do not alter these probabilities. Also, individuals can only move in one direction from compartment to compartment. For example, a recovered individual is assumed to be immune to the disease (i.e they cannot become susceptible or infected again). These assumptions are not met in reality but allow the SIR model to serve as a mathematical baseline for capturing epidemic behavior.

2.2 Derivation of Basic Reproductive Ratio

First, we rewrite equation 2 in the form

$$\frac{dI}{dt} = I(\beta S - \gamma)$$

Assume $S \approx 1$, that is, the entire population is susceptible to a specific disease. For a disease to be able to spread, we require

$$\beta - \gamma > 0$$

$$\frac{\beta}{\gamma} > 1$$

$\frac{\beta}{\gamma}$ is called the basic reproductive ratio (expressed by the symbol R_0). In words, R_0 is the expected number of secondary cases resulting from a primary case in an entirely susceptible population (Keeling & Rohani, 2008).

$R_0 < 1$ corresponds with a disease that will not outbreak, while $R_0 > 1$ means a disease will become an outbreak.

2.3 Simulation of Epidemic Outbreaks

We generated 1500 deterministic SIR outbreaks by numerically solving the SIR ordinary differential equations for each parameter combination. Parameters were sampled independently to reflect a broad range of infectious diseases.

R_0 (basic reproduction number): Sampled uniformly from $[0.5, 5.5]$. This range covers subcritical outbreaks ($R_0 < 1$, where infectious die out) through highly transmissible diseases (e.g., COVID-19, $R_0 \approx 2.5 - 6$). The lower bound of 0.5 allows for modeling non-epidemic scenarios, while the upper bound of 5.5 captures most infectious diseases.

γ (recovery rate): Sampled uniformly from $[1/10, 1/3]$, corresponding to average infectious periods of 3-10 days. This range aligns with data for many infections, from shorter-duration illnesses (e.g., 3 days) to longer-duration diseases (e.g., 10 days). The transmission rate β was derived as $\beta = R_0\gamma$, which follows from the SIR model's relationship.

I_0 (initial infected proportion): Sampled uniformly from $[1e-6, 1e-4]$ (0.0001%-0.01% of the population). This range reflects realistic early outbreak conditions, where only a tiny fraction of a population is initially infected.

Each simulation ran for 20 days, and disease trajectories provided the input for the machine learning models.

2.4 Machine Learning Models

We trained models to predict R_0 from partial epidemic trajectories, treating each simulated trajectory as a feature vector. For each observation window (5, 10, 15, 20 days), the model received the infected proportion at each day as input features.

We compared two regression models:

- Linear Regression (LR): A simple baseline to test if a linear combination of the trajectory can be used for predictive purposes.
- Random Forest Regression (RF): A non-linear method used to capture potential interactions and non-linear relationships.

The dataset of 1,500 simulated outbreaks was split into 80% training (1200 outbreaks) and 20% testing (300 outbreaks). Both models were trained on 5, 10, 15, and 20 days of epidemic trajectories and evaluated on the test set.

Evaluation Metrics:

- R^2 (Coefficient of Determination): Represents the proportion of variation in R_0 that is explained by the epidemic trajectories.
- Mean Squared Error (MSE): Measures the average squared difference between predicted and actual R_0 values.

Primary Analysis: The observation window was the central comparison. We evaluated both models across the four windows to answer: How early in an outbreak can machine learning models estimate R_0 accurately? Results show that predictive performance improves with longer observation windows, with Random Forest outperforming Linear Regression across all windows.

3. Preliminary Results

3.1 Feasibility of Early Estimation

Both models show that R_0 can be estimated from early epidemic trajectories. Predictive performance improves with longer observation windows as shown in Figure 1.

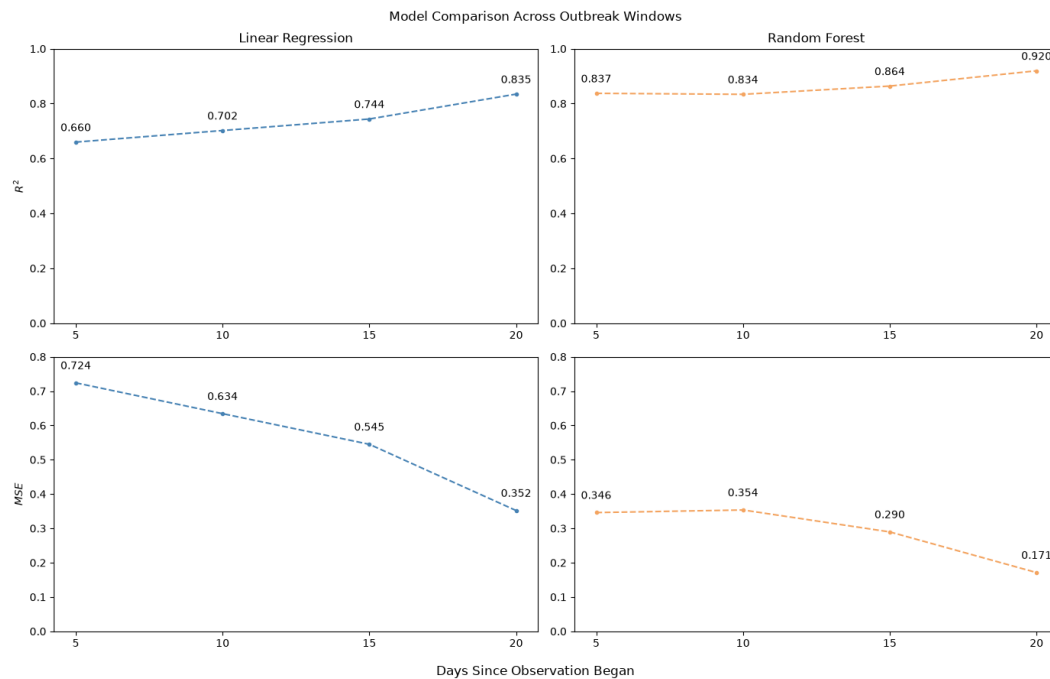


Figure 1: Model Comparison Across Outbreak Windows

Linear Regression and Random Forest show increasing accuracy across the four observation windows, though Random Forest consistently outperforms Linear Regression. This suggests that even partial trajectories provide significant signal for R_0 estimation, addressing the central research question: *How early in an outbreak can machine learning models estimate R_0 accurately?*

3.2 Observation Window Impact

Linear Regression's performance gains are largest from 15 to 20 days. Random Forest only marginally improved across windows, though its baseline predictive performance was already strong ($R^2 = 0.837$). Random Forest's strong performance on the 5-day window indicates that near-optimal estimation of R_0 may be possible within the first week of an outbreak.

3.3 Model Comparison

Random Forest substantially outperforms Linear Regression across all observation windows, as measured by both R^2 and MSE. Further, Random Forest's 5-day window predictive accuracy is higher than Linear Regression's 20-day window. This suggests that non-linear relationships and interactions in the data meaningfully contribute to predictive accuracy. The 20-day window for Random Forest achieved an R^2 of 0.920, meaning that the 20-day epidemic trajectories capture 92% of the variation in R_0 . This is a substantial result, and outperforms Linear Regression's 20-day window by 0.085 (8.5%).

3.4 Residual Analysis

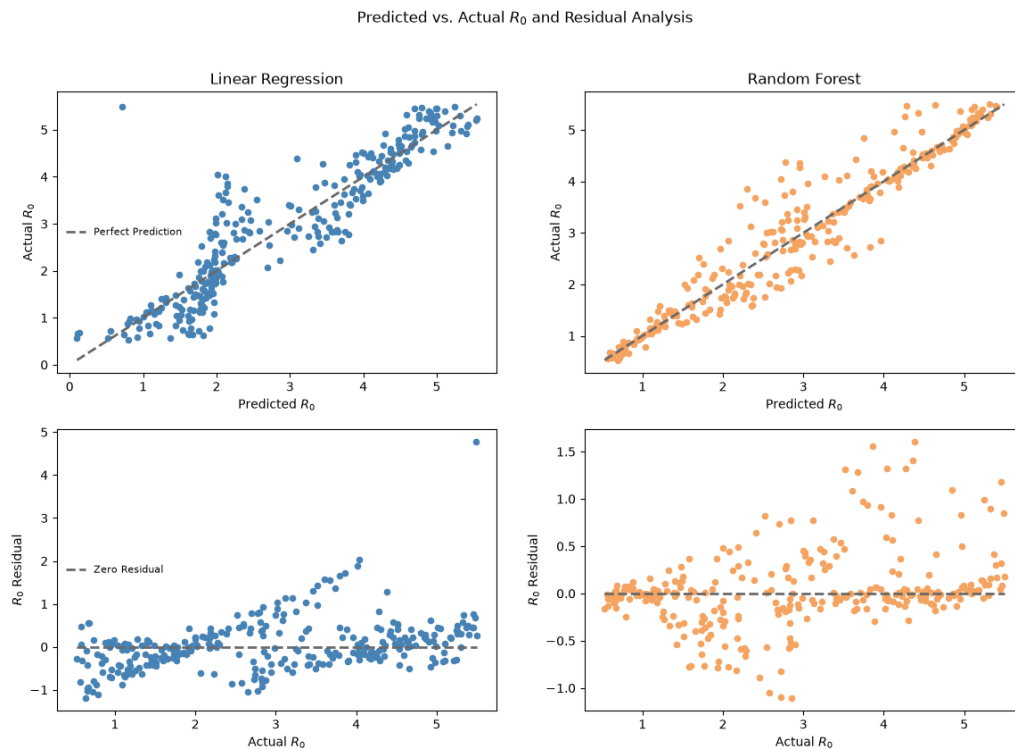


Figure 2: Residual Analysis

Random Forest: Predictions tightly around identity line, but residuals show consistent underestimation for high R_0 values ($R_0 \geq 4$), suggesting a potential limitation in capturing extreme transmission dynamics.

Linear Regression: Residuals are generally farther from zero residual line, further showing its weaker performance relative to Random Forest. There is an extreme outlier (actual $R_0 \approx 5$, predicted $R_0 \approx 1$) that should be further investigated, as it may reveal potential issues with the data pipeline.

These preliminary findings confirm that early R_0 estimation is feasible with machine learning models, and that observation window length is a critical factor in model performance.

4. Discussion

4.1 Early Issues

Although the machine learning models achieved R^2 values approaching unity when trained on static initial infectious proportions, this perfect fit was indicative of **overfitting**. The models had effectively memorized the training trajectories, learning idiosyncratic noise rather than the generalizable relationships between initial conditions and R_0 . This lack of generalization would severely limit their utility for predicting transmission dynamics in real-world epidemic data, where initial conditions are inherently variable.

4.2 Future Work

To address limitations of the current framework and improve real-world application, we will pursue the following extensions as part of a 2-semester honors thesis:

- **Stochastic Simulations:** Introduce variability into epidemic trajectories to allow for better approximation of real-world dynamics. This will test the hypothesis that models trained on stochastic data better generalize to observed outbreaks, even if internal performance metrics (e.g., R^2) decrease.
- **Real-World Data Integration:** We plan to incorporate data from the COVID-19 Open Data Repository and use those outbreaks to train our model. This will allow for hybrid training, where a model can be trained partially on synthetic data as well as real data.
- **Hyperparameter Tuning:** Select Random Forest parameters (e.g., number of trees, tree depth) that minimize MSE, further improving model performance.
- **Expanded Observation Windows:** Explore additional time points (e.g., 25, 30 days) to investigate the point of diminishing returns in predictive performance.
- **Additional Models:** Evaluate other machine learning approaches (e.g., gradient boosting, neural networks) to compare against Linear Regression and Random Forest.

Timeline:

Period	Milestones
Fall 2026	Complete stochastic simulations and begin real-world data integration.
December 2026	Prepare poster abstract submission for ASA symposium.
Spring 2027	Finalize honors thesis with external validation and a comparative analysis. Prepare comprehensive paper for ATP.
April 2027	Finalize formal paper, potentially present at ASA symposium.

This work will serve as a foundation for PhD applications, by demonstrating long-term research ability. I also have plans to submit a poster abstract for the American Statistical Association's Symposium on Data Science in Statistics, which will take place in April, 2027.

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